

K-medoid clustering algorithm steps:

1. Choose K number of random points from the data and assign these K points to K number of cluster. These are the initial medoids.
2. For all the remaining data points, calculate the distance from each medoid and assign it to the cluster with the nearest medoid.
3. Calculate the total cost (sum of all the distances from all the data points to the medoids).
4. Select a random point as the new medoid and swap it with the previous medoid. Repeat 2 & 3 steps.
5. If the total cost of the new medoid is less than that of the previous medoid, make the new medoid permanent and repeat step 4.
6. If the total cost of the new medoid is greater than the cost of the previous medoid, undo the swap and repeat step 4.
7. The Repetitions have to continue until no change is encountered with new medoid to classify data points.

K-medoid algorithm

- ↳ It is an ~~up~~ unsupervised ~~method~~ method with unlabelled data.
- ↳ It is an improved version of the K-Means algorithm with mainly designed to deal with outliers data.
- ↳ It is partitioning algo and compare with other partitioning algo, it is simple, fast and easy to implement.
- ↳ Each cluster must have at least one object.
- ↳ An object must belong to only one cluster.
- ↳ A medoid is a point in the cluster from which the sum of distances to other data points is ~~min~~ minimal.
- ↳ A medoid is a point in the cluster from which dissimilarities with all other points in the clusters are minimal.
- ↳ Thus medoid is taken as reference point in K-medoid algorithm.

k-Medoid Example

i	x	y
x_1	2	6
x_2	3	4
x_3	3	8
x_4	4	7
x_5	6	2
x_6	6	4
x_7	7	3
x_8	7	4
x_9	8	5
x_{10}	7	6

Step 1: ~~Select~~ Select any two random medoids that are random representative of objects:

$$C_1(3, 4), C_2(7, 4)$$

Now make cluster according to these two, for this we make ~~table~~ distance table for each medoid.

Distance formula is as

$$\text{distance} = |a - c| + |b - d|$$

Form distance table for $C_1(3, 4)$.

We remove the these two ~~can~~ medoid from table. and compute the distance of $C_1(3, 4)$ from each point $x_1, x_3, x_4, \dots, x_{10}$ except C_1 & C_2 .

i	x	y	Distance/cost	C_1
x_1	2	6	$ 2-3 + 6-4 =$	3
x_3	3	8	$ 3-3 + 8-4 =$	4
x_4	4	7	$ 4-3 + 7-4 =$	4
x_5	6	2	$ 6-3 + 2-4 =$	5
x_6	6	4	$ 6-3 + 4-4 =$	3
x_7	7	3	$ 7-3 + 3-4 =$	5
x_9	8	5	$ 8-3 + 5-4 =$	6
x_{10}	7	6	$ 7-3 + 6-4 =$	6

Form distance table for center mediod $C_2 (7, 4)$

i	x	y	Distance/cost	C_2
x_1	2	6	$ 2-7 + 6-4 =$	7
x_3	3	8	$ 3-7 + 8-4 =$	8
x_4	4	7	$ 4-7 + 7-4 =$	6
x_5	6	2	$ 6-7 + 2-4 =$	3
x_6	6	4	$ 6-7 + 4-4 =$	1
x_7	7	3	$ 7-7 + 3-4 =$	1
x_9	8	5	$ 8-7 + 5-4 =$	2
x_{10}	7	6	$ 7-7 + 6-4 =$	2

Now compare the $\text{cost}(C_1)$ and $\text{cost}(C_2)$ for every x_i and select the minimum cost.

$\Rightarrow$

C_1	C_2
3	7
4	8
4	6
5	3
3	1
5	1
6	2
6	2

Step II

create the cluster for c_1 & c_2

$$\text{cluster 1} = \{(2,6), (3,8), (4,7), (3,4)\}$$

$$\text{cluster 2} = \{(7,4), (6,2), (6,4), (7,3), (8,5), (7,6)\}$$

Compute total cost:

$$\text{Total cost}(x, c) = \sum_{i=1}^n |x_i - c_i|$$

$$= (3+4+4) + (3+1+1+2+2)$$

$$= 20$$

Step 3: select one of non-medoid 0

$$\text{Let } 0 = \{7, 3\} \text{ i.e. } x_7$$

again do the same procedure as c_2

i	x_i	y_i	Distance/cost	c_2	c_1
x_1	2	6	$ 2-7 + 6-3 = 8$	8	3
x_3	3	8	$ 3-7 + 8-3 = 9$	9	4
x_4	4	7	$ 4-7 + 7-3 = 7$	7	4
x_5	6	2	$ 6-7 + 2-3 = 2$	2	5
x_6	6	4	$ 6-7 + 4-3 = 2$	2	3
x_8	7	4	$ 7-7 + 4-3 = 1$	1	4
x_9	8	5	$ 8-7 + 5-3 = 3$	3	6
x_{10}	7	6	$ 7-7 + 6-3 = 3$	3	6

Now compare the result c_1 & c_3 and select the minimum cost

$$\therefore c_1 = (3), (4), (4), 5, 3, 4, 6, 4$$

$$c_3 = 8 \quad 9 \quad 7 \quad (2) \quad (2) \quad (1) \quad (3) \quad (3)$$

$$\text{Total cost} = 3+4+4+2+2+1+3+3 = 22$$

Step 4: So cost of swapping medal from c_2 to c_3

$S = \text{current total cost} - \text{past total cost}$

$$= 22 - 20$$

$$= 2 > 0$$

So moving c_3 could be bad idea so previous choice was good.